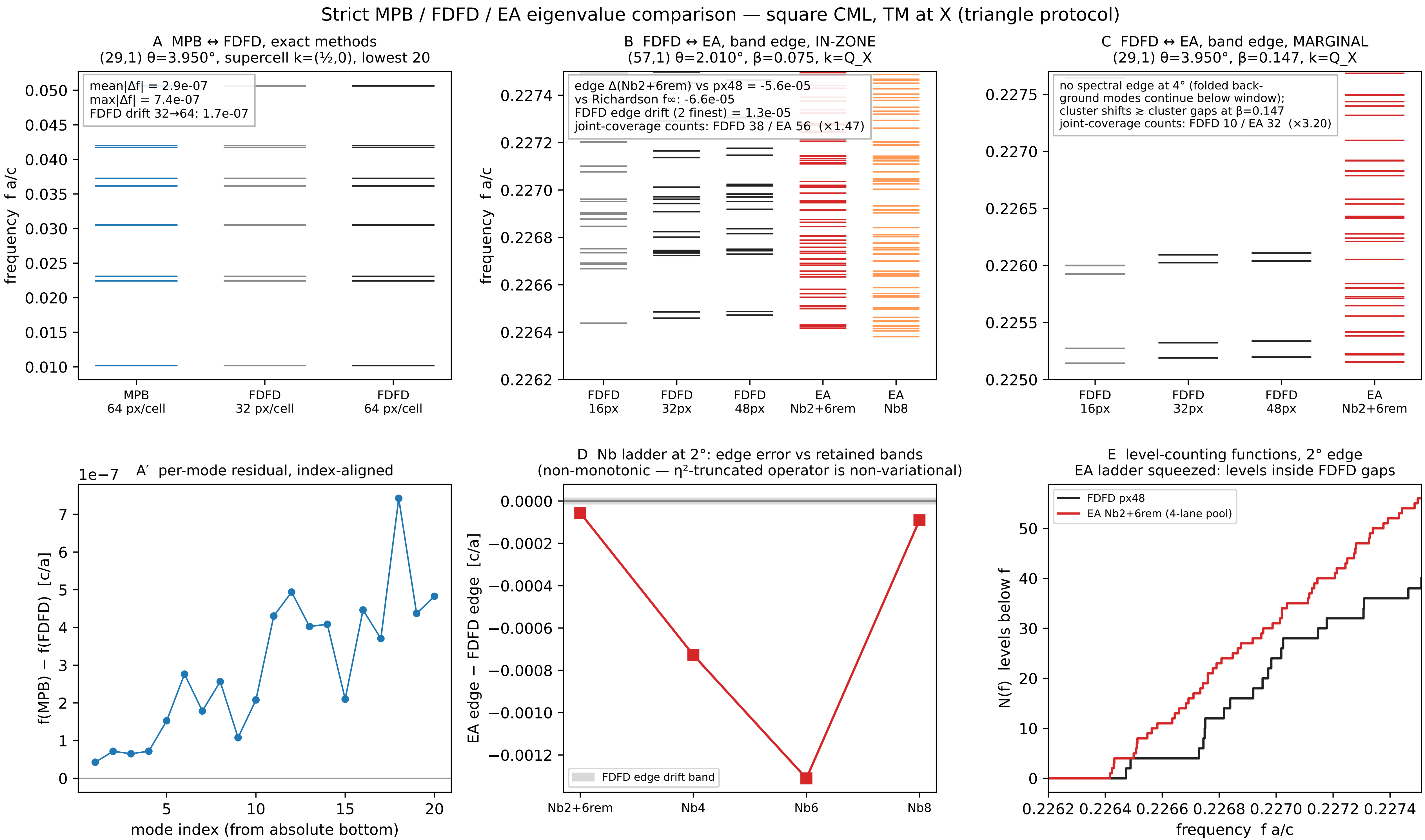


```
Square lattice (const a), TM (Ez) | rods r/a=0.2, ε=8.9 in ε_bg=1.0 | twisted bilayer, both layers in-plane
(57,1): θ=2.0102°, L=57.009a, 3250 cells | (29,1): θ=3.9499°, L=29.017a, 842 cells | accuracy zone β=θ_rad/γ, γ(X,0→1)=0.468: 2°→β=0.075 (in-zone), 4°→β=0.147 (marginal)
EA: blaze2d V4 exact-TM (Berry-free), k₀ = monolayer X = (½,0)·2π/a, band 0, λ_ref=2.35584 (f_ref=0.24428) | registry 128², envelope N_s=128², fd_order 4 | shift-invert f₀=0.2270, 15 modes/lane
EA k-lanes: Γ_m,X1_m,X2_m,M_m (2×2 CML fold), pooled, no valley doubling | FDFD: L=ε^{-1/2}(Σ g^{ab}D†D)ε^{-1/2}, subpixel ε-avg N_sub=8, CHOLMOD shift-invert, tol 1e-10
FDFD bottom runs: Q_X=(π,0) ≡ supercell k=(½,½), σ_ω=0.2270 | MPB: res 64/cell, 20 bands, run_tm, supercell k=(½,0) — FDFD leg A matches MPB's k exactly
STRICT protocol: sorted ladders, no Hungarian/assignment matching; panel A index-aligned from absolute mode 1
(m,1) supercell = 2× CENTERED cell: τ=(L1+L2)/2 ∈ both layers' lattices (verified exactly) → FDFD ×2 pairs = primitive-
cell folding = X⊕X′ valleys at Q_X; EA lanes {Γ_m,M_m} = valley X, {X1_m,X2_m} = valley X′ (C4-paired, verified to 2e-5)
```